DESIGN TOOLS GUIDE

BONUS 3











Bead Pattern Design Tools Guide

Design your own seed-bead patterns with graph paper, a spreadsheet and a few friendly apps | Tiny Bead Studio

Most beaders start by following other people's patterns, and that is a perfectly good way to learn. But sooner or later a quiet question shows up: what if I could make my own? A bracelet in the exact colors of your kitchen, earrings that match a favorite scarf, a motif nobody else is selling. Designing your own charts is the step that turns bead weaving from following into creating.

The good news is that you do not need special talent or expensive software. A sheet of graph paper and a pencil will carry you a long way. A free spreadsheet will carry you further. And a handful of dedicated bead-pattern tools can save time once you know what you want from them.

This guide walks you through the whole path in small steps: why custom patterns matter, the paper and spreadsheet methods, the main tools worth knowing about, how to build a symmetric motif, how to turn a simple photo or shape into a chart, how to match your chart to the real bead colors on your table, and how to export and print a pattern you can actually bead from. It closes with eight practice exercises and a list of the mistakes almost everyone makes once.

A note on software: apps change. Prices, platforms and features are updated all the time, and some tools come and go. Wherever this guide names a tool, treat the description as a starting point and check the current version, price and system requirements before you download or buy anything. The paper and spreadsheet methods will work forever.

1. Chapter 1: Why Design Your Own Patterns

When you buy or download a pattern, you are borrowing someone else's decisions: the colors, the width, the length, the shape of the motif. When you design your own, you make those decisions yourself, and that brings several practical benefits.

- Perfect fit: you choose the exact length and width for your wrist, your neck or your ear, instead of resizing someone else's chart.
- Your colors: you build around the beads you already own, which saves money and uses up stash.
- Originality: a pattern that is truly yours is easier to present as your own work if you later decide to sell finished pieces.
- Understanding: once you have drawn a chart, you read every other chart faster, because you know why it is built the way it is.
- Confidence: the first time a piece you designed comes off the loom or needle exactly as you pictured it is a genuine milestone.

There is also a quiet business reason. Patterns you design are patterns you fully control. You do not have to wonder whether a shared chart may be reused for products you sell. That said, rules about copying other people's designs vary, so when you are inspired by someone else's work, change it enough to make it your own, and check the terms that come with any pattern you buy.

> **START SMALL**
> Your first original design does not need to be impressive. A 10-bead-wide bracelet with a simple repeating diamond is a real design. Small wins build the habit.

Before you open any tool, it helps to know the three things every bead chart is made of: a grid that matches how the beads sit, a set of colors with a clear key, and a size that fits the finished piece. Everything else is decoration. The rest of this guide is about making those three things easy.

2. Chapter 2: The Graph-Paper and Spreadsheet Method

Paper first. Plain graph paper works for loom and square stitch, where beads line up in tidy rows and columns, just like the squares on the page. For peyote and brick stitch, the beads are offset by half a bead from row to row, so square paper is a stand-in only, or you can use dedicated offset paper that you print yourself. Many beaders simply chart peyote on a square grid and remember that every other row is shifted.

Delica 11/0 beads are cylinders, roughly 1.6 mm wide, and a little shorter than they are wide, so each real bead is slightly wider than tall. On paper this makes very little difference for small patterns. If a design needs to look exactly right, such as a portrait or a letter, choose a grid with cells that are a little wider than tall, or check how the piece looks once a few rows are woven.

> **QUICK SIZING RULE**
> Delica 11/0 beads laid side by side run about 16 to the inch (check your own beads, since brands and lots vary slightly). So a 7-inch bracelet is roughly 112 beads long, and a bracelet 12 beads wide is about three quarters of an inch wide.

Now the spreadsheet method, which works in Google Sheets (free with a Google account) or Excel. The idea is simple: every cell is one bead, and the fill color of the cell is the bead color.

- Make the cells small and nearly square: select all, set columns to a width of about 22 to 28 pixels and rows to a similar height. In Google Sheets, use the Resize column and Resize row options.
- Turn off the extra clutter: hide the formula bar and, if you wish, keep the gridlines visible, since they are your bead outlines.
- Decide your width first. Mark the left and right edges with a thicker border so you always know where the piece ends.
- Fill cells with the paint-bucket tool. Pick a handful of colors, put them in a small key at the side, and stick to them.
- Add a row number column on the left and a column number row on top. You will thank yourself later at the loom or stitching table.
- Use copy and paste for repeats. Select a finished motif, copy it, and paste it down the page for a border or a repeating bracelet pattern.

Task	Google Sheets	Excel
Resize cells to squares	Select cells, right-click, choose Resize column and Resize row	Select cells, Format menu, set Column Width and Row Height
Fill a cell with color	Fill color button in the toolbar	Fill Color button on the Home tab
Copy a motif	Select, Ctrl or Cmd C, then V	Select, Ctrl or Cmd C, then V
Mirror a motif	Copy, paste, then redraw reversed (see Chapter 4)	Same; Excel can also flip with formulas, but redrawing is simpler
Print to scale	File, Print, set scale and margins	Page Layout, Scale to Fit, then Print

Symmetry in a spreadsheet is simplest if you draw one half, copy it, and redraw it in reverse beside

3. Chapter 3: Free and Low-Cost Tools Worth Knowing

Dedicated bead software can do things a spreadsheet cannot do so smoothly: show a woven preview, count beads automatically, and switch between stitch styles. The tools below exist and are commonly mentioned by beaders. Features and prices change, so for each one, check the current version, price and system requirements on the maker's own page before you download.

Tool	What it is good for	Things to check
Google Sheets	Free grid charts, copy and paste, sharing a chart on phone or computer. Best first tool.	Needs a free Google account; cells must be resized to squares.
Microsoft Excel	Same idea as Sheets, with stronger printing control.	Paid or part of a subscription; check your version.
JBead	A desktop program built for bead weaving: draws charts for loom, peyote and brick-style layouts, shows a simulation of the finished piece and a color list.	Check the current version, the operating systems supported and whether it needs Java installed.
BeadTool 4	Desktop software for designing seed-bead patterns, with stitch-specific views and bead-count lists.	Check current price, trial availability and whether it runs on your computer.
Canva	Making a polished printable version of a chart, with a title, a color key and your studio name. Not a bead-chart program by design.	Free and paid plans exist; check what is included.
Procreate	Drawing on an iPad with a grid or guides; good for freehand motifs you then convert to charts.	Paid app for iPad only; check the current price.

Here is a way to think about choosing among them. If you are just starting, use graph paper and Google Sheets until you have made three or four designs. That teaches you what you actually need from software. Then, if you find yourself changing designs often or wanting to see a woven preview before you bead, try a dedicated tool such as JBead or BeadTool 4, using a free trial if one is offered.

> **A CAUTION ABOUT DOWNLOADS**
>
> Only download software from the maker's own website or from a store you trust. Avoid files shared on random forums. If an offer sounds too good, it is usually not worth the risk to your computer.

What about the many web tools and apps that turn a picture into a bead chart? They exist and can be fun, but quality varies and so do their rules and prices. Treat the result as a rough draft: the colors will need editing, stray single beads need cleaning up, and the final chart should always be checked by eye before you weave it.

Whichever tool you choose, keep one habit: save a plain copy of every finished chart, either as an image or a PDF, in a folder named after the piece and the date. Software changes; a PDF never stops opening.

Practice task: tool comparison

Choose two tools from the table. Make the same small design, a 10-bead-wide diamond border 30 rows long, in each one. Write down how long each took, what you liked, and what annoyed you. Keep the one that felt better.

4. Chapter 4: Designing a Symmetric Motif Step by Step

Symmetry is the shortcut to looking professional. A motif that mirrors itself feels balanced, and it is easier to design because you only draw half. Here is a clear method that works on paper, in a spreadsheet, or in any bead program.

- Choose an odd width for the piece, such as 11 or 13 beads, so there is a single center column. This gives the motif a spine and avoids an awkward split down the middle.
- Draw the center column first. Mark it lightly or color it as a guide.
- Design the left half, working outward from the center. Use only two or three colors for your first motif.
- Copy the left half to the right side as a mirror image. Mirror means the bead that was three columns left of center is now three columns right of center, in the same row.
- Check the rows. Each row should read the same from the left edge to the center as from the right edge to the center.
- Add a one-bead border of a contrasting color around the outside, if you like. Borders make edges neat and hide small irregularities.
- Step back and squint. If you can see the motif clearly even when blurred, the contrast is good. If it fades, make the main shape darker or lighter.

To mirror in a spreadsheet, say your center column is column G. A bead in column D sits three to the left of center, so its partner sits in column J. Copy the color across by hand, or use the paint-bucket tool for each matching cell. Some beaders also use a small helper formula to copy colors, but redrawing is often just as quick and easier to understand.

Motif type	Why it works	Good first use
Diamond	Simple stepped shape, easy to repeat	Bracelet borders
Zigzag	Two colors, a clear rhythm	Bands and cuffs
Flower	Needs a center bead and symmetrical petals	Earring or pendant
Heart	Mirror with a rounded top; needs careful row counting	Gift-set pendant
Geometric stripe	Zero drawing skill needed	Beginner loom bracelet

> **THE REPEAT RULE**
> If a bracelet is made of repeating motifs, make sure the motif's height divides your total length neatly. A 14-row motif repeated 8 times gives 112 rows, about 7 inches in Delica 11/0. Count before you design the details.

Finally, check the beginning and end. A repeating pattern should look finished where it starts and where it stops, so plan a partial or full border at both ends. It is easy to forget this and end up with a design that looks cut off.

Practice task: a symmetric diamond

On graph paper, make a piece 11 beads wide. Draw a central diamond about 9 beads tall, then mirror it. Add a one-bead border. Choose two colors for the diamond and one for the background. Count how many beads of each color

5. Chapter 5: Turning a Photo or Shape into a Bead Chart

Turning an image into a chart is really an act of simplifying. A bead chart is a very low-resolution picture. A 12-bead-wide bracelet is only 12 pixels across. Think of it as tiny pixel art, and your job is to decide what to keep.

- Start with a simple image: a silhouette, a leaf, a heart, an initial, a single flower. Complex photos with a lot of detail do not translate into small bead charts.
- Make it high contrast. In any photo editor, raise the contrast and reduce the number of colors to three or four.
- Reduce the size to your chart width. If your bracelet is 12 beads wide, the image should be 12 pixels across. Many free editors can resize an image this small, though it will look blurry.
- Use the small image as a guide. Place it beside your spreadsheet and copy cell colors one by one, or use a tool that converts images to a grid automatically.
- Clean up. Delete lone single beads of one color that float on their own. They look like mistakes when woven.
- Weave a test strip. Before you commit to the whole piece, bead a short sample of the tricky area to see how it reads in beads.

A shape tracing method works well with a pencil. Print your image at the size of your chart, lay graph paper over it or draw the grid over it, and shade each square that is mostly filled by the shape. Squares that are half filled need a decision: include them if the shape would look too thin without them, leave them out if the shape would look too heavy.

> KEEP IT LEGIBLE
>
> Letters and numbers need at least 5 beads of height to be readable, and 7 beads is more comfortable. Names on bracelets are better done on a wider piece.

As for rights, only use images you have the right to use. Your own photos, drawings and simple shapes are the safest sources. If you draw inspiration from a copyrighted illustration or a brand logo, redraw loosely in your own style, or better, leave it alone and design something original.

Practice task: heart from an image

Find or draw a simple heart. Resize it to 11 pixels wide. Convert it to a two-color chart (heart and background). Clean up any stray beads and weave a sample on a small loom or a few rows of peyote.

6. Chapter 6: Matching Your Chart to Real Bead Colors

Screens lie about color. A turquoise on your monitor can look gray-blue in your hand, and the same bead can look different under daylight and a lamp. The solution is to design with real beads in mind from the start.

- Work from the real beads. Lay out the beads you own in good daylight and choose your colors from them.
- Name colors in your chart with a code or a short description, such as C1 for cream, C2 for teal, C3 for gold. Add the manufacturer's color number if the tube has one.
- Know the finishes. Matte, opaque, transparent, metallic and lined beads behave differently side by side. A glossy bead next to a matte one reads as different even if both are red.
- Keep contrast in mind. Two colors that look different on screen may look almost identical when woven. Pick colors that differ in lightness, not only in hue.
- Keep the palette small. Three to five colors is plenty for a first design. More colors mean more thread changes and more chances of a muddy result.

A tip that saves money: before buying a new color, sew one or two beads onto a scrap of felt or tape a few to a card, and place them beside the beads they will sit next to. Many beaders keep a small swatch card for every color they own, with the number and the brand.

Palette idea	Colors	Mood
Soft neutral	Cream, taupe, light gold	Calm, everyday wear
Ocean	Navy, teal, white	Fresh, summery
Berry	Plum, rose, silver	Elegant
Earth	Rust, olive, cream	Warm, autumn
Monochrome	Three shades of one color	Modern and easy to match

Keep a color key beside the chart that lists each symbol or color with the number of beads required. Add 10 to 15 percent for spares, since you will drop some. If a color is discontinued or sold out, a clear key makes swapping easy.

> **CHECK BEFORE ORDERING**
> Colors and stock change. Before ordering a large amount of one color, buy a small tube first and test it, and check the label or the supplier's current listing.

Practice task: build a swatch card

Take a plain piece of cardstock. Tape or sew one bead of each color you own, with its code written beside it. Choose a 4-color palette from the card and write it as a key for your next design.

7. Chapter 7: Exporting, Printing and Reading Your Chart

A chart is only useful if you can read it comfortably while you bead. That means large enough cells, clear symbols, row numbers and good lighting. Here is a simple checklist for turning a draft into a finished, printable pattern.

- Use large cells. Cells that are too small are tiring to read. Aim for a size where each cell is comfortably visible at arm's length.
- Number rows and columns. For a loom piece, numbering from the bottom up matches how you weave. For peyote, many beaders number from the starting end.
- Add a color key with the number of beads per color.
- Mark the center column and the repeat boundaries with a bold line or a small arrow.
- Add the finished size in inches and millimeters, plus the bead type and approximate count. For example: 12 beads wide, 112 rows, about 0.75 by 7 inches, Delica 11/0.
- Include the stitch. A chart for peyote is read differently from a chart for loom, so write it at the top.
- Split long charts onto two pages with a few rows of overlap, labeled clearly.

To export from a spreadsheet, either print directly to PDF or use the Download option and choose PDF. Set the page to portrait for tall charts, remove margins that waste space, and check the preview before printing. In Canva you can place a screenshot or exported image of your chart on a page with a title and your studio name. Always save a copy as a PDF.

Format	Good for	Watch out for
PDF	Printing and sharing, never changes	Check that it is not resized when printed
PNG image	Placing in documents or on a phone	Blurry if enlarged
Native tool file	Editing later in the same software	Only opens in that program
Paper original	Quick notes while you bead	Easy to lose, so scan or photograph it

> **TAPE AND TICK**
> Many beaders tape the chart to a clipboard and tick off each row as they finish it with a pencil or sticky marker. A magnetic board with a ruler strip also works well.

Last, test your pattern by actually making it. A chart that looks perfect on screen can still hide an error: a row that is one bead short, a color that disappears. Make the piece once, note corrections on the chart, and only then consider it final.

Practice task: a finished pattern sheet

Take one of your charts and turn it into a one-page pattern sheet with a title, finished size, bead list with counts, stitch, color key and the chart itself. Print it and ask yourself: could someone else bead this from the page alone?

8. Chapter 8: Eight Design Exercises and Common Mistakes

Skills grow with repetition. These eight exercises build on each other. Do one per session and you will have a small portfolio of original charts within a couple of weeks.

Practice task: Exercise 1: The stripe

Design a 10-bead-wide, 40-row piece of horizontal stripes in three colors. Learn how color order changes the mood.

Practice task: Exercise 2: The zigzag

Design a 12-bead-wide chevron pattern that repeats every 8 rows. Check that the repeat meets itself cleanly.

Practice task: Exercise 3: The diamond border

Design a 13-bead-wide piece with a centered diamond and a one-bead border. Mirror it carefully.

Practice task: Exercise 4: Initial letter

Design a single initial 7 beads tall on a contrasting background, with a border.

Practice task: Exercise 5: Tiny flower

Design a flower motif 9 beads across, with a center bead, four to six petals and a leaf. Use no more than four colors.

Practice task: Exercise 6: Gradient

Design a 12-bead-wide band that shifts gradually from a dark shade to a light shade in five steps.

Practice task: Exercise 7: Earring pair

Design a 6-bead-wide, 20-row earring drop in two colors. Make sure both earrings match exactly in orientation.

Practice task: Exercise 8: From photo to chart

Take a simple silhouette, convert it to a chart, clean it up, and make a full printable pattern sheet.

Now for the mistakes almost every designer makes once. Knowing them in advance saves a lot of unpicked thread.

- Too many colors. Start with three. Add a fourth only if the design clearly needs it.
- Even width for a symmetric design. This leaves no center column and makes mirroring awkward.
- Forgetting the edges. Without a border, the edge beads can look ragged.
- Ignoring bead size differences. Different brands of 11/0 beads vary slightly; test in the exact beads you plan to use.
- Tiny details. A single bead of contrast in a large field often looks like a mistake instead of a feature.
- No test piece. Always weave a short sample, especially for new motifs.
- Poor contrast. Two similar tones disappear when woven.
- No saved copy. Keep a PDF and a plain image of every final chart.

8. Chapter 8: Eight Design Exercises and Common Mistakes (continued)

> **A GENTLE RULE**
>
> Finish a design before you start a new one. Unfinished charts pile up quickly, and a finished imperfect piece teaches more than a perfect idea never beaded.

Next steps

You now have a complete, practical path from a blank page to a finished, printable bead pattern. Start with the grid, keep your palette small, mirror your motifs, test your colors with real beads, and always weave a sample before you commit.

Remember that tools are only helpers. The best software in the world will not replace the experience of beading a design and seeing what works. Make one small original piece this week, even a very simple one, and keep a copy of the chart. That is the beginning of your own pattern collection.

Next step: choose one exercise from Chapter 8, set aside an hour, and chart it. Happy designing.